SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Somasundhari.M(196Signature / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Somasundhari. M.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	2	1	1	1	1
Debugging	25%	1	2	1	1	1
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

24/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

1. Given Configuration

Parameter	**Configured value**
IP address	192.168.10.65
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1
Required subnet	192.168.10.0/26

2. Identify the Subnet

The subnet mask is 255.255.255.192.

Block size:

\[256-192=64 \]

Therefore, the subnet ranges are:

- 192.168.10.0 – 192.168.10.63
- 192.168.10.64 – 192.168.10.127
- 192.168.10.128 – 192.168.10.191
- 192.168.10.192 – 192.168.10.255

The configured IP address 192.168.10.65 belongs to the second subnet, 192.168.10.64/26, not the required subnet 192.168.10.0/26.

3. Network Address, Broadcast Address and Valid Host Range

For the required subnet 192.168.10.0/26:

Particular	**Address**
Network address	192.168.10.0
First valid host	192.168.10.1
Last valid host	192.168.10.62
Broadcast address	192.168.10.63

For the configured IP's actual subnet, 192.168.10.64/26:

Particular	**Address**
Network address	192.168.10.64
First valid host	192.168.10.65
Last valid host	192.168.10.126
Broadcast address	192.168.10.127

4. Verify the IP Address and Default Gateway

IP address: 192.168.10.65

- It is a valid host address in the subnet 192.168.10.64/26.
- It does not belong to the required subnet 192.168.10.0/26.
- Therefore, it is incorrectly configured for the required network.

Default gateway: 192.168.10.1

- It is a valid host address in 192.168.10.0/26.
- However, it is outside the workstation's configured subnet, 192.168.10.64/26.

- Therefore, the gateway is inconsistent with the workstation's current IP configuration. Normally, the default gateway must be reachable on the workstation's local subnet.

5. Identify the Configuration Errors

The errors are:

1. The workstation's IP address belongs to the wrong subnet.
2. The default gateway belongs to a different subnet from the configured IP address.
3. The workstation cannot communicate directly with systems in 192.168.10.0/26 because its configured IP places it in 192.168.10.64/26.
4. The IP address and gateway must be configured consistently with the required subnet.

6. Corrected Network Configuration

To connect the workstation to the required subnet, use the following configuration:

Parameter	**Corrected value**
IP address	192.168.10.2
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1
Network address	192.168.10.0
Broadcast address	192.168.10.63

The IP address 192.168.10.2 is an example of a suitable host address, assuming it is not already assigned to another device. The gateway 192.168.10.1 must be configured on the router or Layer 3 device connected to this subnet.

With this correction, the workstation and gateway belong to the same subnet, allowing local communication when the physical connections and other network settings are correct.

7. Calculate the Total Number of Usable Host Addresses

For a /26 subnet:

$$\text{Host bits}=32-26=6$$

Total IP addresses:

\[2^6=64 \]

Usable host addresses:

\[2^6-2=64-2=\boxed{62} \]

Therefore, the corrected subnet supports 62 usable host IP addresses.

8. Summary

Requirement	Result
Original IP's subnet	192.168.10.64/26
Required subnet	192.168.10.0/26
Required valid host range	192.168.10.1–192.168.10.62
Required broadcast address	192.168.10.63
Is original IP suitable?	No
Is original gateway valid for the original subnet?	No
Corrected workstation IP	192.168.10.2
Corrected gateway	192.168.10.1
Usable host addresses after correction	62

9. Conclusion

The communication problem occurs because the workstation's IP address and default gateway belong to different /26 subnets. Assigning the workstation an unused IP address such as 192.168.10.2, while retaining the correct subnet mask and gateway 192.168.10.1, resolves the addressing mismatch. The corrected subnet provides 62 usable host addresses.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

1. Given Configuration

Parameter	**Device A**	**Device B**
IPv6 address	2001:db8::ff00:42:8329/64	2001:db8:0:0:1::1/64
Prefix length	/64	/64

IPv6 addresses are 128 bits long and consist of eight groups of hexadecimal digits. The /64 prefix indicates that the first 64 bits represent the network prefix.

2. Expand Both IPv6 Addresses

Device A

Given address: 2001:db8::ff00:42:8329/64

The :: represents four zero groups.

Expanded address:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B

Given address: 2001:db8:0:0:1::1/64

The :: represents three zero groups.

Expanded address:

2001:0db8:0000:0000:0001:0000:0000:0001/64

3. Determine the Network Prefix of Each Device

For a /64 prefix, the first four groups (64 bits) identify the network.

Device A:

Expanded address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix:

2001:0db8:0000:0000/64

Compressed form:

2001:db8::/64

Device B:

Expanded address:

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix:

2001:0db8:0000:0000/64

Compressed form:

2001:db8::/64

4. Verify Whether Both Devices Belong to the Same Subnet

Compare the first four groups of both addresses:

Device	First 64 bits (network prefix)
Device A	2001:0db8:0000:0000
Device B	2001:0db8:0000:0000

Both devices have the same network prefix, 2001:db8::/64.

Result: Yes, both devices belong to the same IPv6 subnet.

5. Identify the Addressing or Prefix Mismatch

There is no IPv6 subnet-prefix mismatch in the given configuration.

The fifth group in Device B's address is 0001, but because the prefix length is /64, this group belongs to the interface identifier, not the network prefix.

Therefore, the different interface identifiers do not prevent communication between devices on the same subnet.

If communication still fails, possible causes include:

- Incorrect physical or network-interface configuration.
- IPv6 disabled on one of the devices.
- Firewall rules blocking traffic.
- Duplicate IPv6 addresses.
- Incorrect VLAN configuration or other Layer 2 connectivity problems.

6. Optimize the IPv6 Configuration

Since both devices already belong to the same /64 subnet, their IPv6 addresses do not need to be changed to resolve a subnet mismatch.

The following configuration can be retained:

Parameter	**Device A**	**Device B**
IPv6 address	2001:db8::ff00:42:8329/64	2001:db8:0:0:1::1/64
Network prefix	2001:db8::/64	2001:db8::/64
Status	Valid address format	Valid address format

Both addresses are valid within the same subnet. The administrator should check firewall settings, interface status, VLAN connectivity, and duplicate-address detection to identify the actual cause of the communication failure.

For a simplified configuration, the administrator could assign Device B another unused address in the same subnet, such as 2001:db8::2/64, provided it is not already assigned. This is an optional change, not a necessary correction.

7. Calculate the Number of Available Host Addresses

IPv6 addresses contain 128 bits.

Given prefix length:

\[/64 \]

Number of interface identifier bits:

\[128-64=64 \]

Total possible addresses in the subnet:

\[2^{64} \]

\[=\boxed{18,446,744,073,709,551,616} \]

Therefore, the subnet contains \(2^{64}\) possible IPv6 addresses. IPv6 does not use broadcast addresses, so the IPv4 rule of subtracting two addresses does not apply.

8. Summary Table

Requirement	**Result**
Device A expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329/64
Device B expanded address	2001:0db8:0000:0000:0001:0000:0000:0001/64
Device A network prefix	2001:db8::/64
Device B network prefix	2001:db8::/64
Same subnet?	Yes
Prefix mismatch?	No
Address correction required?	No
Possible addresses in subnet	\(2^{64}\)

9. Conclusion

Both devices have the same network prefix, 2001:db8::/64, and therefore belong to the same IPv6 subnet. The given addresses do not contain a subnet-prefix mismatch. The communication failure must be investigated through other network settings, such as firewall rules, interface configuration, or VLAN connectivity. The subnet contains \(2^{64}\) possible IPv6 addresses.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

1. Introduction

Subnetting divides a large network into smaller networks called subnets. However, using the same or unnecessarily large subnet sizes for departments can waste IP addresses, while small subnets may not provide enough host addresses.

Variable Length Subnet Masking (VLSM) allows different subnet masks to be assigned according to each department's requirements, improving IP address utilization.

The organization has been allocated the network 192.168.1.0/24 and uses /26, /27, and /28 subnet masks.

2. Given Data

Subnet mask	**Subnet mask in decimal**	**Total IP addresses**	**Usable host addresses**
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

The original /24 network contains \(2^8=256\) IP addresses, from 192.168.1.0 to 192.168.1.255.

Since the question does not specify the number of departments or their exact host requirements, the following is an example of a valid, non-overlapping allocation using all three prefix lengths.

3. Calculate the Network Address, Host Range and Broadcast Address

A. First subnet – /26

Subnet mask: 255.255.255.192

Block size:

\[256-192=64 \]

- Network address: 192.168.1.0
- First valid host: 192.168.1.1
- Last valid host: 192.168.1.62
- Broadcast address: 192.168.1.63
- Usable host addresses: \(2^6-2=\boxed{62}\)

B. Second subnet – /27

Subnet mask: 255.255.255.224

Block size:

\[256-224=32 \]

The next available subnet starts at 192.168.1.64.

- Network address: 192.168.1.64
- First valid host: 192.168.1.65
- Last valid host: 192.168.1.94
- Broadcast address: 192.168.1.95
- Usable host addresses: \(2^5-2=\boxed{30}\)

C. Third subnet – /28

Subnet mask: 255.255.255.240

Block size:

\[256-240=16 \]

The next available subnet starts at 192.168.1.96.

- Network address: 192.168.1.96
- First valid host: 192.168.1.97
- Last valid host: 192.168.1.110
- Broadcast address: 192.168.1.111
- Usable host addresses: \(2^4-2=\boxed{14}\)

4. Summary of Subnet Allocation

Particulars	**Subnet 1**	**Subnet 2**	**Subnet 3**
Prefix	/26	/27	/28
Network address	192.168.1.0	192.168.1.64	192.168.1.96
Valid host range	.1–.62	.65–.94	.97–.110
Broadcast address	.63	.95	.111
Total IP addresses	64	32	16
Usable host addresses	62	30	14

These three subnets do not overlap.

5. Calculate Allocated and Unused IP Addresses

Total addresses in the original /24 network:

\[2^{32-24}=256 \]

Total addresses allocated to the three subnets:

\[64+32+16=112 \]

Remaining unallocated addresses:

\[256-112=\boxed{144} \]

The remaining unallocated range is:

192.168.1.112 – 192.168.1.255

6. Identify Inefficient Subnet Allocations

The following problems may occur:

1. Oversized subnet: A department needing only 10 hosts does not need a /26, which supports 62 usable hosts.
2. Undersized subnet: A department requiring 35 hosts cannot use a /27, because it supports only 30 usable hosts.
3. Wasted host capacity: Assigning a subnet much larger than the department's actual requirement leaves usable addresses unassigned.
4. Overlapping subnets: If two subnets are assigned overlapping address ranges, routing and communication problems may occur.
5. Poor address planning: Assigning smaller subnets before larger ones can complicate allocation if the available address space becomes fragmented.

7. Debugging the Addressing Plan

The administrator should verify the following:

- Each subnet starts at a valid network boundary for its prefix.
- The broadcast address of one subnet does not overlap the next subnet.
- Every device has an IP address within its assigned subnet's valid host range.
- The subnet mask configured on each device matches its department's subnet.
- The number of usable host addresses is sufficient for the department's current requirements.

For example, assigning 192.168.1.95 to a host in the /27 subnet would be an error because .95 is that subnet's broadcast address.

8. Recommend an Optimized VLSM Scheme

VLSM assigns subnet sizes according to the number of hosts required by each department. The administrator should first identify the largest department and allocate its subnet, followed by smaller departments.

For example, if the department requirements were 50, 25, and 12 hosts, respectively, the following allocation would be suitable:

Department (example)	Required hosts	Recommended prefix	Usable hosts
Department A	50	/26	62
Department B	25	/27	30
Department C	12	/28	14

This uses 112 total addresses and meets all three example requirements. The network ranges calculated in Sections 3 and 4 can be retained for this allocation.

Important: The actual host requirements are not supplied in the question, so this is an illustrative VLSM plan rather than a confirmed optimization for the organization's real departments. A truly optimized allocation requires the actual host count for each department, including expected growth.

9. Conclusion

The /24 network contains 256 addresses. The example /26, /27, and /28 allocations use 112 addresses and leave 144 addresses unallocated for future expansion. VLSM improves address utilization by assigning subnet sizes based on departmental needs. By checking host requirements, network boundaries, broadcast addresses, and subnet masks, the administrator can reduce waste and prevent addressing errors.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet

1. Given IPv6 Addresses

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

  The prefix length /64 means the first 64 bits (first four hextets) represent the network prefix.

  2. Expand Both IPv6 Addresses

  IPv6 addresses contain eight hextets, each consisting of four hexadecimal digits.

  Device A:

  2001:db8::ff00:42:8329/64

  Expanded notation:

  2001:0db8:0000:0000:0000:ff00:0042:8329/64

  Device B:

  2001:db8:0:0:1::1/64

Expanded notation:

2001:0db8:0000:0000:0001:0000:0000:0001/64

3. Determine the Network Prefix

For a /64 prefix, the first four hextets identify the network.

Details	Device A	Device B
Network prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Interface ID	0000:ff00:0042:8329	0001:0000:0000:0001

Both devices have the same first four hextets.

4. Verify Whether Both Devices Belong to the Same Subnet

Yes. Both devices belong to the same subnet:

2001:db8::/64

Therefore, there is no network prefix mismatch in the given configuration. Both addresses are valid IPv6 unicast addresses, and their interface identifiers are different.

5. Identify the Possible Configuration Error

Since both devices belong to the same /64 subnet, the communication failure is not caused by a subnet mismatch.

Possible causes include:

- Incorrect IPv6 configuration on a device.
- Firewall rules blocking IPv6 traffic.
- Devices connected to different VLANs.
- Duplicate IPv6 addresses.
- Network interface or link problems.

The router, switch, and device configurations should be checked to identify the actual cause.

6. Optimize the IPv6 Configuration

Use addresses from the same /64 subnet and ensure each device has a unique address.

Device	Optimized IPv6 address
Device A	2001:db8::1/64
Device B	2001:db8::2/64
Network prefix	2001:db8::/64

These addresses are in the same subnet and can communicate directly when the interfaces, VLANs, and security settings are configured correctly.

7. Calculate the Number of Available Host Addresses

IPv6 address length = 128 bits.

Network prefix = 64 bits.

Host/interface identifier bits:

\[128-64=64\text{ bits} \]

Number of possible interface identifiers:

\[2^{64}=18,446,744,073,709,551,616 \]

Therefore, the /64 subnet supports \(2^{64}\) possible interface identifiers. Unlike conventional IPv4 subnet calculations, IPv6 does not use the rule of subtracting two addresses for the network and broadcast addresses.

Conclusion

Both original IPv6 addresses belong to the same 2001:db8::/64 subnet, so there is no prefix mismatch. Assigning unique addresses such as 2001:db8::1/64 and 2001:db8::2/64 provides a clear configuration for testing. If communication still fails, the VLAN, interface, firewall, and routing settings must be checked.

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5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

1. Given Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route (0.0.0.0/0)	10.0.0.1

Destination IP: 192.168.2.100

2. Identify the Matching Route Using Longest-Prefix Match

The longest-prefix match rule selects the route with the most matching network bits.

The destination IP 192.168.2.100 is checked against the routing table:

- 192.168.1.0/24 — No match.
- 192.168.2.0/24 — Match.

- 0.0.0.0/0 — Matches any IPv4 destination, but is less specific.

Therefore, the selected route is 192.168.2.0/24.

3. Determine the Next-Hop Router

The matching routing entry is:

Parameter	Value
Destination IP	192.168.2.100
Matching network	192.168.2.0/24
Next-hop address	10.0.0.3

Hence, the router forwards the packet to 10.0.0.3.

4. Verify the Destination Subnet

The network 192.168.2.0/24 has subnet mask 255.255.255.0.

- Network address: 192.168.2.0
- First usable host: 192.168.2.1
- Last usable host: 192.168.2.254
- Broadcast address: 192.168.2.255

Since 192.168.2.100 falls within the usable host range, it belongs to the intended subnet.

Conclusion: The destination IP and the routing entry are correct.

5. Identify Possible Routing Errors

Because the matching route is correct, the failure may be caused by other configuration issues:

1. Next-hop reachability: Router 10.0.0.3 may be unreachable.
2. Missing return route: The destination network may not have a route back to the sender.
3. Interface failure: The outgoing interface may be down.
4. Firewall or ACL: Security rules may block the packets.
5. Incorrect downstream routing: The next-hop router may lack a route to 192.168.2.0/24.

These should be checked before changing the destination route.

6. Optimize the Routing Table

The routing table can be maintained as follows:

Destination Network	Next Hop	Purpose
192.168.1.0/24	10.0.0.2	Route to Network 1
192.168.2.0/24	10.0.0.3	Route to Network 2
0.0.0.0/0	10.0.0.1	Route for all other destinations

Optimization steps:

- Keep the specific /24 routes for the intended networks.
- Ensure each next hop is reachable through an active interface.
- Remove duplicate or obsolete routes, if any exist.
- Configure return routes on downstream routers.
- Use route summarization only when the networks are contiguous and share a valid common prefix.

The current table is logically correct; no route change is required based on the information provided.

7. Identify the Default Route

The default route is:

0.0.0.0/0 → 10.0.0.1

It is used when no more specific routing entry matches the destination IP.

For example, a packet destined for 8.8.8.8 would use the default route, assuming no more specific route exists.

Conclusion

The destination 192.168.2.100 belongs to 192.168.2.0/24. Using longest-prefix matching, the router selects next hop 10.0.0.3. Since the routing entry is correct, connectivity, return routing, interfaces, and firewall rules should be checked. The default route 0.0.0.0/0 forwards packets for destinations without a more specific matching entry.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated

Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation